R5.3: √élu with a three-prime NTT

Round 5, 2026-10-18. Code `examples/r5_3_ntt_sqrt_velu.rs` and `src/cryptanalysis/isogeny_eval/ntt.rs` (native Rust). Raw records: `threshold.json`, `rational.json`, `generic_karatsuba.json`, `generic_ntt.json`; fits and derived table: `analysis.json` / `analysis.txt`. Full write-up: `RESULTS.md`. Toy, synthetic instances only; no sub-rho or deployed-curve claim.

Question and hypothesis

Round 4 measured √élu at exponent 0.82 [0.805, 0.838] with Karatsuba. Hypothesis: with NTT products the exponent drops toward 0.5–0.6; the rational-kernel conclusion only strengthens; the generic crossover moves only if kernel finding speeds up.

Method

Same $j = -3375$ vertical instances (generator ported unchanged; 87/87 rational and 16/16 generic rows reproduce round-4 primes, codomains and find/ladder counts). Unit: M = $\mathbb{F}_p$ multiplication; W = word product mod a 62-bit NTT prime, **stated** at 1 M; I = 91 M. Under `Auto` the library takes the NTT only where its counted cost is below Karatsuba's for that exact shape.

Results (measured)

series	range	exponent [95% CI]
√élu Karatsuba	$2^{16}..2^{24}$	0.822 [0.805, 0.838]
√élu NTT	$2^{16}..2^{24}$	0.798 [0.775, 0.820]
√élu NTT	$2^{20}..2^{24}$	0.745 [0.686, 0.804]
generic find, Karatsuba	31..863	2.587 [2.550, 2.625]
generic find, NTT	31..863	2.582 [2.544, 2.620]

At $\ell = 2^{24}$ the NTT arm counts 2.21e7 Meq vs 2.45e7 (ratio 1.11).

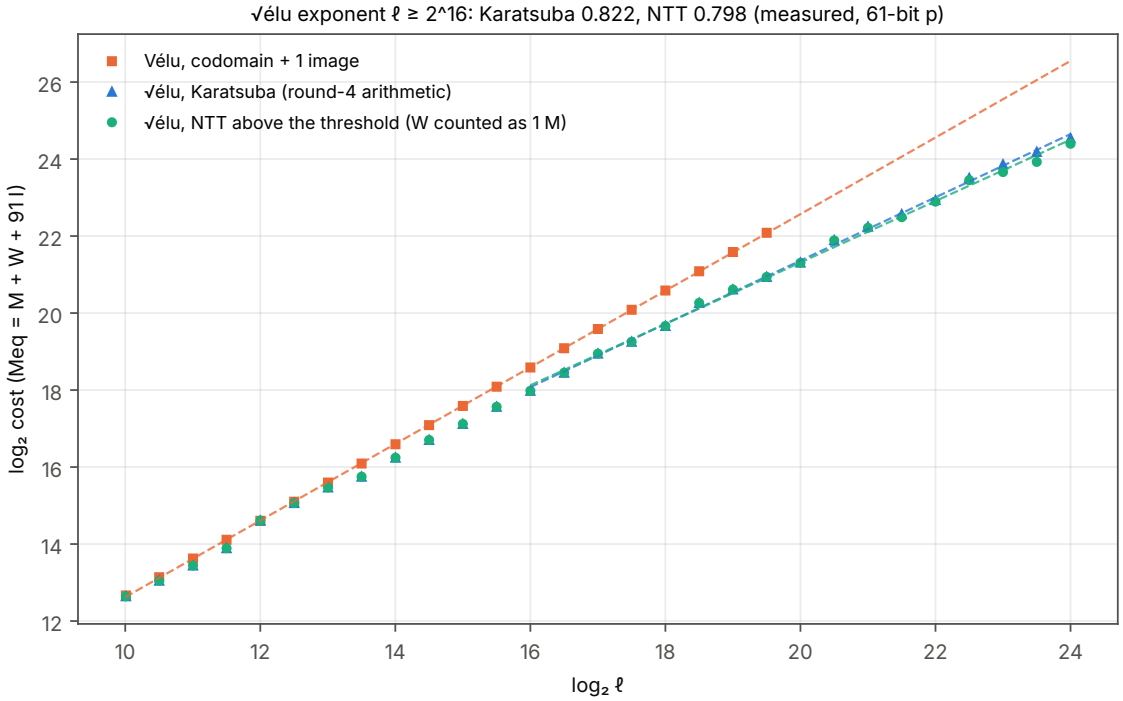


Figure 1: √élu counted cost against ℓ, both arms (measured).

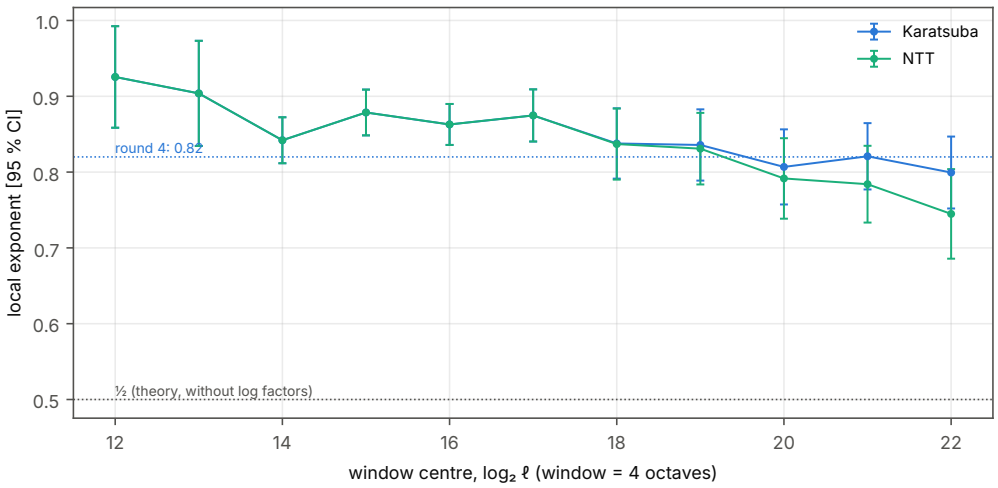


Figure 2: Local exponents over 4-octave windows (measured fits).

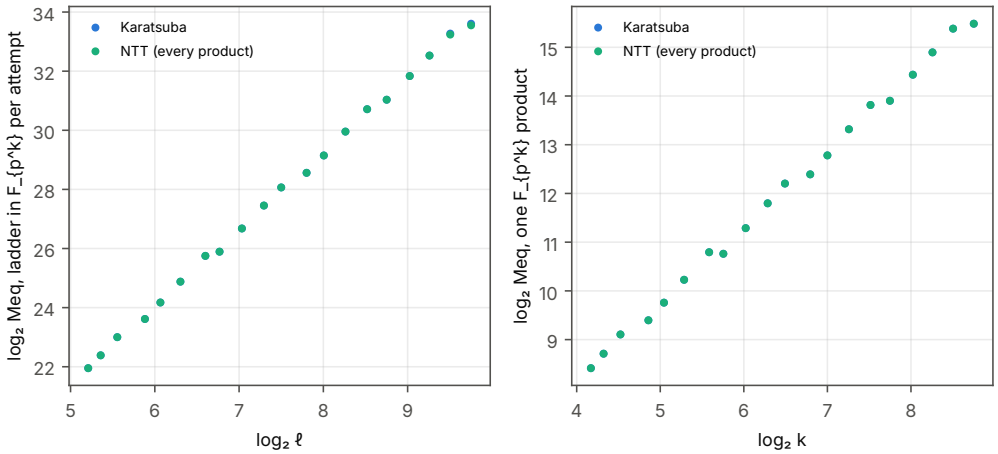


Figure 3: Generic kernel finding, both arms (measured).



Figure 4: NTT vs Karatsuba counted cost and contended wall time per product shape.

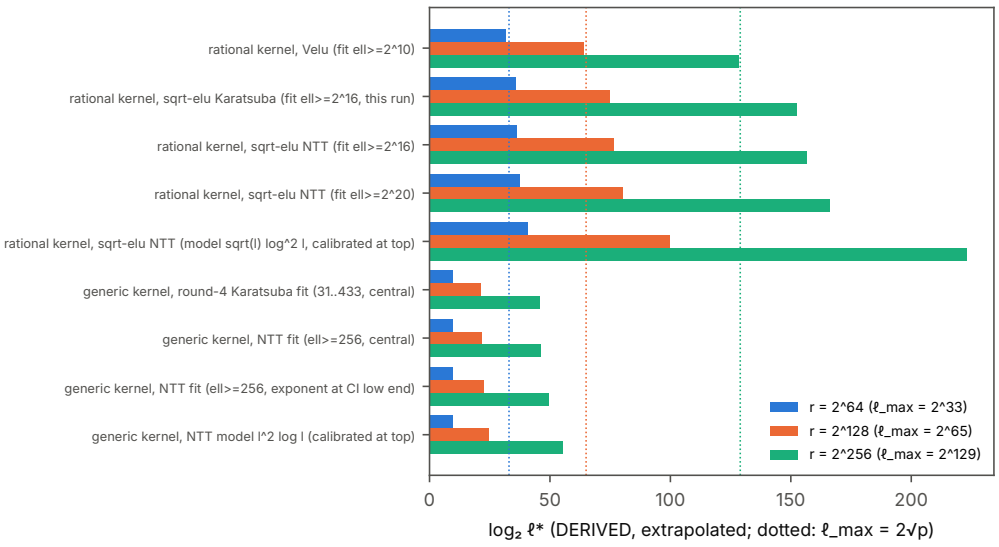


Figure 5: Derived (extrapolated) crossovers with rho.

Derived crossovers ($g = 6$, log2 ℓ^*)

model	$r = 2^{64}$	$r = 2^{128}$	$r = 2^{256}$
$\ell_{\max} = 2\sqrt{p}$	33	65	129
rational, √élu NTT fit ≥ 2^{16}	36.4	76.5	156.8
rational, √élu NTT fit ≥ 2^{20}	37.4	80.4	166.3
generic, round-4 fit	9.6	21.5	45.7
generic, NTT fit	9.6	21.7	46.4

Verdict

Exponent toward 0.5–0.6: **refuted at the measured range** (trend downward, log factors dominate up to 2^{24}). Rational-kernel conclusion: **confirmed and stronger** (no crossover within $\ell \le 2\sqrt{p}$). Generic crossover: **unchanged** ($2^{21.5}$–2^{25} at $r = 2^{128}$), since kernel finding did not speed up.

Limits

W = 1 M is stated, not measured; contended wall shows this NTT costlier per unit. Generic arm capped at $\ell = 863$. Crossovers are extrapolations.